

Hovercraft Hockey Hackathon

Build a puck that moves by barely touching the floor.

Win condition: Best glide plus control in a gym-safe tournament.

THE SCIENCE YOU ARE USING

Air cushion cuts friction. A balloon pushes air down through a hole in a disc, lifting the disc on a thin cushion of air. With almost no contact, friction nearly vanishes, so a small push sends it gliding. Newton's laws then keep it moving until something stops it.

Glide versus control. Maximum glide is easy; control is hard. You balance lift and steerability to compete in gym-safe target hockey. Build quality and a clear explanation also score.

HOW TO BUILD AND RUN IT

- 1 Mount the valve.** Glue a pop-top cap over the center hole of the disc, sealing the edge.
- 2 Attach the balloon.** Stretch an inflated balloon over the closed cap so air escapes only through the hole.
- 3 Test the glide.** Open the cap, set it down, and give a gentle push. Tune for a long, smooth glide.
- 4 Practice control.** Learn to start and stop on target. Too much lift means no control.
- 5 Play target hockey.** Compete in gym-safe rounds. Glide and control both count.

Safety: Latex allergy note. No face-level launches. Play floor-only. Allergy: Latex balloon alternative: small bags or school-approved nonlatex balloons.

MATERIALS

Recycled CDs or discs	per team
Pop-top bottle caps	per team
Balloons	per team
Hot glue	staff station
Targets	shared
Floor tape	shared

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why):

The ONE variable we are testing:

Baseline result:

After redesign:

DEFEND YOUR DESIGN

What evidence made you change your design?

If you ran it again, what is the ONE thing you would change?

HOW YOU SCORE (100 POINTS)

Scoring criterion	Points
Glide performance	30
Target competition score	25
Explanation	20
Redesign	15
Build quality	10
Total	100